1 Neurons

All state is fixed-point and updated *lazily*: a compartment only advances when an event touches it, decaying from its last-event timestamp τ to the current time t. Time is the `u16` event clock (ticks).

1.1 Base-2 decay operator

Every leak is the same $O(1)$ base-2 approximation of an exponential (`math::decay`):

$$\mathcal{D}_k(v, \Delta t) = v \cdot 2^{-\Delta t/2^k}, \tag{1}$$

i.e. v halves every 2^k ticks (half-life 2^k). The exponent k selects the time constant of each compartment; the same operator decays alpha, branch voltage, and the soma potential.

1.2 Synapse

Let $x \in$ `u8` (position along the dendrite), $w \in$ `i8` (weight), $\alpha \in$ `u8` (activity / eligibility trace), $\tau \in$ `u16` (last-event time).

$$s = (x,\ w,\ \alpha,\ \tau) \tag{2}$$

Synapses on a dendrite are ordered by strictly increasing, unique x. The trace α decays with $k_\alpha = 11$ and is boosted by $\alpha_b = 64$ each time a forward action potential arrives:

$$\alpha \leftarrow \mathcal{D}_{k_\alpha}\big(\alpha,\ t - \tau\big), \qquad \alpha \leftarrow \min(\alpha + \alpha_b,\ 255). \tag{3}$$

1.3 Basal dendrite

Synapses more distal than i (larger x) gate i's contribution: their (decayed) traces sum, with weight that falls off over x with $k_x = 4$. This makes integration order-dependent.

$$\gamma_i = \sum_{j:\,x_j > x_i} \mathcal{D}_{k_x}\big(\alpha_j,\ x_j - x_i\big) = \sum_{j:\,x_j > x_i} \alpha_j\, 2^{-(x_j - x_i)/2^{k_x}}. \tag{4}$$

A synaptic event at synapse i carrying presynaptic burst b (the number of APs) leaks the branch voltage V_B since its last event, then adds the gated EPSP ($k_B = 9$):

$$V_B \leftarrow \mathcal{D}_{k_B}\big(V_B,\ \Delta t\big) \ + \ b\, w_i\, (1 + \gamma_i). \tag{5}$$

The basal branch is a hard threshold: on crossing θ_B it emits a dendritic spike and resets.

$$V_B \geq \theta_B \implies \text{dendritic spike}, \quad V_B \leftarrow 0. \tag{6}$$

1.4 Apical dendrite

The apical branch integrates V_B identically (with $k = 11$) but *does not* reset — it leaks. Instead of a discrete spike it delivers a graded plateau to the soma, a sigmoidal transfer (Payeur et al., 2021) with half-activation θ_B, ceiling δV_S and slope $\kappa = \ln 2/2^k$:

$$\sigma^{\mathrm{ap}}(V_B) = \frac{\delta V_S}{1 + e^{-\kappa\,(V_B - \theta_B)}}, \qquad \sigma^{\mathrm{ap}} \in [0,\ \delta V_S]. \tag{7}$$

1.5 Dendrite → soma coupling

A basal dendritic spike with branch constant c delivers $v_S = \max(c, 1)$ to the soma. Distal branches ($c \leq 0$) attenuate to unity at the soma but reinforce their own active synapses locally:

$$v_S = \max(c,\ 1), \qquad \alpha_j \leftarrow \alpha_j + |c| \quad (\text{for } \alpha_j > H_\alpha). \tag{8}$$

An apical event delivers $v_S = \sigma^{\mathrm{ap}}(V_B)$.

1.6 Soma

The soma holds potential $V_S \in \mathrm{i8}$ and a 6-bit burst counter $\beta \in [0, 63]$. An incoming v_S leaks the potential ($k_S = 10$) and the counter ($T_\beta = 500$ ticks per unit) over Δt, then integrates:

$$V_S \leftarrow \mathcal{D}_{k_S}\big(V_S,\, \Delta t\big) + v_S, \qquad \beta \leftarrow \beta - \left\lfloor \tfrac{\Delta t}{T_\beta} \right\rfloor. \tag{9}$$

If the threshold $\theta_S > 0$ is crossed the soma emits a somatic spike whose burst n counts how many multiples of threshold accrued; it then resets to $V_{\text{reset}} = -32$ and the burst reinforces β:

$$V_S \geq \theta_S \implies n = \left\lfloor \tfrac{V_S}{\theta_S} \right\rfloor, \quad V_S \leftarrow V_{\text{reset}}, \quad \beta \leftarrow \min(\beta + n,\ 63). \tag{10}$$

1.7 Plasticity (back-propagating AP)

A somatic spike drives burst-dependent plasticity across the neuron's own afferent synapses. Only sufficiently active synapses ($\alpha_i > H_\alpha$, with $H_\alpha = 30$) update; the burst counter sets the sign about the baseline $H_\beta = 4$ ($\beta > H_\beta$: LTP; $\beta < H_\beta$: LTD), and η is the per-neuron learning rate ($\eta \geq 120$):

$$\Delta w_i = \frac{(\beta - H_\beta)\,\alpha_i}{\eta}, \qquad w_i \leftarrow \mathrm{clamp}\big(w_i + \Delta w_i,\ -127,\ 127\big). \tag{11}$$

1.8 Event flow

One spike threads a fixed sequence of events, the burst count b riding the payload throughout:

$$\texttt{SYNAPSE\_SIGNAL} \rightarrow \begin{cases} \texttt{DENDRITIC\_SPIKE} \rightarrow \texttt{SOMA\_SIGNAL} & \text{(basal)} \\ \texttt{SOMA\_SIGNAL} & \text{(apical)} \end{cases} \rightarrow \texttt{SOMATIC\_SPIKE} \rightarrow \texttt{SYNAPSE\_SIGNAL}. \tag{12}$$

A somatic spike fans out to every axonal target as an independent `SYNAPSE_SIGNAL`.